CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 17: Special Types of Quadrilaterals

Topic-wise short notes with solved examples

The trapezium, parallelogram, rectangle, rhombus, square and kite: their definitions, properties of sides, angles and diagonals, and problems based on them.

1 Quadrilaterals

- A **quadrilateral** has 4 sides, 4 angles, 4 vertices and 2 diagonals.
- The **sum of its angles is** $360°$.
- **Adjacent sides** meet at a vertex; **opposite sides** do not. **Adjacent angles** share a side; **opposite angles** do not.

Example 1.1 The angles of a quadrilateral are in the ratio $3 : 4 : 5 : 6$. Find them.

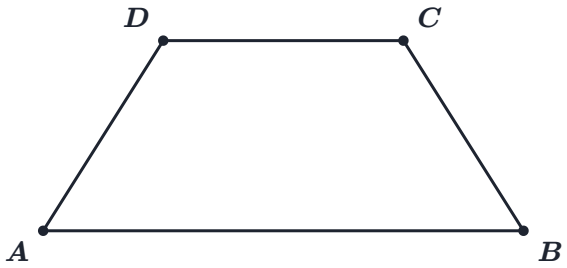
SOLUTION

$18x = 360°$, so $x = 20°$

$60°, 80°, 100°, 120°$

2 Trapezium

- A **trapezium** has **one pair of opposite sides parallel**.
- The two angles on each non-parallel side add up to $180°$ (co-interior angles).
- An **isosceles trapezium** has equal non-parallel sides. Its base angles are equal and its diagonals are equal.



Isosceles trapezium: AB is parallel to DC and AD = BC

Example 2.1 In trapezium ABCD, $AB \parallel DC$, $\angle A = 65°$ and $\angle B = 80°$. Find $\angle C$ and $\angle D$.

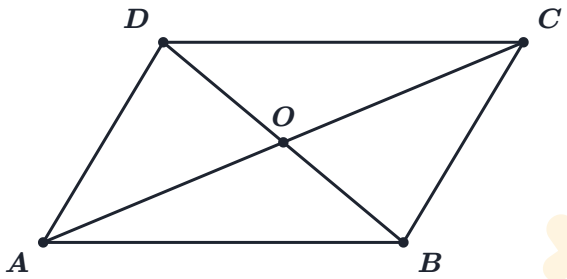
SOLUTION

$\angle D = 180° - 65°$ and $\angle C = 180° - 80°$

$\angle C = 100°$, $\angle D = 115°$

3 Parallelogram

- A **parallelogram** has **both pairs of opposite sides parallel**.
- Opposite sides are **equal**.
- Opposite angles are **equal**.
- Adjacent angles are **supplementary** (add up to $180°$).
- The diagonals **bisect each other**.
- Each diagonal divides it into two congruent triangles.



Parallelogram: OA = OC and OB = OD

Example 3.1 In parallelogram ABCD, $\angle A = 70°$. Find the other angles.

SOLUTION

$\angle C = 70°$, $\angle B = \angle D = 110°$

Example 3.2 Two adjacent sides of a parallelogram are in the ratio $3 : 5$ and its perimeter is 64 cm. Find its sides.

SOLUTION

$2(3x + 5x) = 64$, so $x = 4$

12 cm and 20 cm

Example 3.3 In parallelogram PQRS, $\angle P = (2x + 15)°$ and $\angle R = (3x - 20)°$. Find x and all the angles.

SOLUTION

Opposite angles are equal: $2x + 15 = 3x - 20$, so $x = 35$.

$\angle P = \angle R = 85°$, $\angle Q = \angle S = 95°$

4 Rectangle

- A **rectangle** is a parallelogram in which **every angle is** $90°$.
- It has all the properties of a parallelogram.
- Its diagonals are **equal** and bisect each other, so the four half-diagonals are equal.

Example 4.1 The diagonals of rectangle ABCD meet at O. If $\angle AOB = 70°$, find $\angle OAB$ and $\angle OBC$.

SOLUTION

$OA = OB$, so $\angle OAB = \dfrac{180° - 70°}{2} = 55° = \angle OBA$. Then $\angle OBC = 90° - 55°$.

$55°$ and $35°$

Example 4.2 A rectangle has sides 9 cm and 12 cm. Find the length of a diagonal.

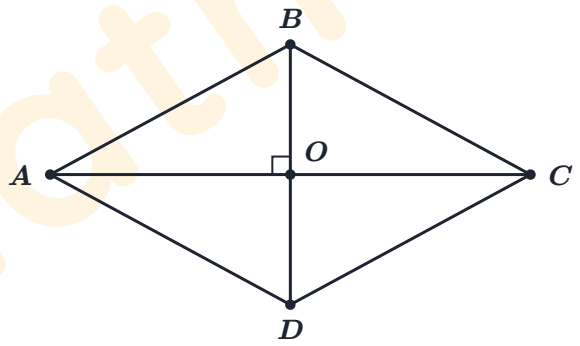
SOLUTION

$\sqrt{81 + 144}$

15 cm

5 Rhombus

- A **rhombus** is a parallelogram with **all sides equal**.
- It has all the properties of a parallelogram.
- Its diagonals **bisect each other at right angles**.
- Each diagonal **bisects the angles** at the vertices it joins.
- The diagonals are not equal (unless it is a square).



Rhombus: the diagonals bisect each other at right angles

Example 5.1 The diagonals of a rhombus are 16 cm and 12 cm. Find its side and perimeter.

SOLUTION

Half-diagonals 8 and 6: side $= \sqrt{64 + 36} = 10$ cm

10 cm; 40 cm

Example 5.2 In rhombus ABCD, $\angle ABC = 56°$. Find $\angle ACD$.

SOLUTION

$\angle BCD = 180° - 56° = 124°$. Diagonal AC bisects it.

$62°$

6 Square

- A **square** is a rectangle with all sides equal; it is also a rhombus with every angle $90°$.
- All sides are equal and every angle is $90°$.
- The diagonals are **equal**, **bisect each other at right angles**, and make $45°$ with the sides.
- Diagonal of a square of side a is $a\sqrt{2}$.

Example 6.1 The diagonal of a square is $7\sqrt{2}$ cm. Find its side and area.

SOLUTION

7 cm; 49 sq cm

7 Kite

- A **kite** has **two pairs of equal adjacent sides**: $AB = AD$ and $CB = CD$.
- The diagonal AC (the axis of symmetry) is the **perpendicular bisector** of BD.
- AC bisects $\angle A$ and $\angle C$. The angles at B and D are equal.

Example 7.1 In kite ABCD with $AB = AD$ and $CB = CD$, $\angle A = 100°$ and $\angle C = 60°$. Find $\angle B$.

SOLUTION

$\angle B = \angle D$, so $2\angle B = 360° - 160°$

$100°$

8 Comparing the diagonals

Property of the diagonals	Parallelogram	Rectangle	Rhombus	Square	Kite
Bisect each other	Yes	Yes	Yes	Yes	No
Equal	No	Yes	No	Yes	No
Perpendicular	No	No	Yes	Yes	Yes
Bisect the angles	No	No	Yes	Yes	One only

- Every square is a rectangle and a rhombus. Every rectangle and every rhombus is a parallelogram. Every parallelogram is a trapezium in the sense of having a pair of parallel sides.

9 Common mistakes

- Taking adjacent angles of a parallelogram as equal instead of supplementary.
- Saying the diagonals of a parallelogram are equal.
- Saying the diagonals of a rectangle are perpendicular.
- Using the full diagonals instead of half-diagonals to find the side of a rhombus.
- Forgetting that the diagonals of a rhombus bisect its angles.
- Treating a kite as a parallelogram.

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